

Comprehensive Statistics Terms & Regression Diagnostics Glossary

This document serves as an in-depth reference glossary for key statistical concepts, hypothesis testing procedures, and regression diagnostics, including formulas, component breakdowns, and intuitive conceptual explanations for study and revision.

1. Hypothesis Testing Framework

- **Plain-English Definition:** A structured inferential method used to decide whether sample data provides sufficient evidence to reject a baseline claim about a population parameter in favor of an alternative effect.
- **Core Formulas:**

$$\text{Test Statistic} = (\text{Point Estimate} - \text{Null Value}) / \text{Standard Error}$$

Decision Rule: Reject H_0 if $p\text{-value} \leq \alpha$ (or if $|\text{Test Statistic}| \geq |\text{Critical Value}|$)

- **What Each Term Represents:**
 - **Point Estimate:** The sample statistic calculated directly from data (e.g., sample mean $\bar{x}$ or difference in means $\bar{x}_1 - \bar{x}_2$).
 - **Null Value:** The hypothesized value of the population parameter under the null hypothesis H_0 (frequently 0, representing no effect or zero difference).
 - **Standard Error (SE):** The standard deviation of the sampling distribution, quantifying random sampling variation.
 - **p-value:** The probability of observing a test statistic as extreme as, or more extreme than, the observed value, assuming H_0 is true.
 - **α (Significance Level):** The threshold for Type I error tolerance, standardly set at 0.05 (5%).
- **What I Understood (Core Concept):** You assume the default state of the world is "nothing is happening" or there is no difference (H_0). The test statistic measures how many standard errors away your observed sample result lies from that baseline. A tiny p-value (< 0.05) reveals that under H_0 , your sample would be exceptionally rare; hence, you reject H_0 . One never "accepts" or "proves" the alternative, but merely concludes the evidence strongly contradicts H_0 .

2. Shapiro-Wilk Test

- **Plain-English Definition:** A formal goodness-of-fit hypothesis test used to verify whether a sample of continuous data is drawn from a normally distributed population.
- **Formula:**

$$W = [\sum(a_i * x_{(i)})]^2 / [\sum(x_i - \bar{x})^2]$$

- **What Each Term Represents:**
 - **W:** The Shapiro-Wilk test statistic (bounded between 0 and 1; values close to 1 indicate normality).
 - **$x_{(i)}$:** The i-th order statistic (sample values sorted in ascending order: $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$).
 - **$\bar{x}$:** The arithmetic sample mean.
 - **a_i :** Precalculated weights derived from the means, variances, and covariances of order statistics from a standard normal distribution.
 - **$\sum(x_i - \bar{x})^2$:** Total sum of squared deviations from the sample mean.
- **What I Understood (Core Concept):** It acts as a gatekeeper test. Standard parametric tests (such as Student's t-test) rely on normality. The Shapiro-Wilk test correlates your sorted data against the ideal theoretical values of a bell curve. If $W \approx 1$ and $p > 0.05$, the normality assumption holds. If $p < 0.05$, the bell curve assumption fails, necessitating non-parametric methods (like Mann-Whitney) or mathematical transformations.

3. Levene's Test (Homogeneity of Variance)

- **Plain-English Definition:** An inferential test to assess the assumption of equal population variances (homoscedasticity) across two or more independent groups.
- **Formula:**

$$W = [(N - k) / (k - 1)] * [\sum n_i * (\bar{Z}_{.i} - \bar{Z}_{..})^2 / \sum \sum (Z_{ij} - \bar{Z}_{.i})^2]$$

- **What Each Term Represents:**
 - **Z_{ij} :** Absolute deviation $|Y_{ij} - \bar{Y}_i|$ (or median-based $|Y_{ij} - \tilde{Y}_i|$ for Brown-Forsythe) of observation j in group i from its group center.
 - **$\bar{Z}_{.i}$:** Mean absolute deviation within group i.
 - **$\bar{Z}_{..}$:** Grand mean of all absolute deviations across the entire dataset.
 - **k:** Total number of distinct groups.
 - **n_i :** Sample size of group i.
 - **N:** Total combined sample size across all groups ($N = \sum n_i$).
- **What I Understood (Core Concept):** Many parametric comparisons assume that while group means may differ, their dispersion (spread) is consistent. Levene's test calculates each data point's distance from its group center and performs an ANOVA on those distances. A p-value < 0.05 indicates significantly unequal variances, requiring adjustments such as Welch's t-test.

4. Two-Sample t-Test (Independent Samples)

- **Plain-English Definition:** A parametric hypothesis test comparing the population means of two distinct, unrelated groups.
- **Formulas:**

Equal Variance (Pooled):

$$t = (\bar{X}_1 - \bar{X}_2) / [s_p * \sqrt{(1/n_1 + 1/n_2)}]$$

$$\text{where } s_p^2 = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2] / (n_1 + n_2 - 2)$$

Unequal Variance (Welch's t-test):

$$t = (\bar{X}_1 - \bar{X}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$$

- **What Each Term Represents:**
 - $\bar{X}_1, \bar{X}_2$: Sample means of group 1 and group 2.
 - s_1^2, s_2^2 : Sample variances of group 1 and group 2.
 - n_1, n_2 : Sample sizes of group 1 and group 2.
 - s_p^2 : Weighted pooled variance combining both groups.
 - t : Calculated test statistic compared against Student's t-distribution.
- **What I Understood (Core Concept):** The test takes the raw difference between the sample averages and standardizes it by the estimated variability of that difference. It evaluates whether the observed gap between groups is substantial or simply the result of random sampling noise. If $|t|$ is large and $p < 0.05$, the two populations are concluded to have distinct means.

5. Mann-Whitney U Test (Wilcoxon Rank-Sum)

- **Plain-English Definition:** A non-parametric test evaluating whether one group's values systematically rank higher or lower than another's, without assuming normality.
- **Formulas:**

$$U_1 = n_1 n_2 + [n_1(n_1 + 1) / 2] - R_1$$

$$U_2 = n_1 n_2 - U_1$$

$$U = \min(U_1, U_2)$$

- **What Each Term Represents:**
 - n_1, n_2 : Sample sizes of group 1 and group 2.
 - R_1 : Sum of ranks assigned to group 1 after ranking all combined observations from 1 to $(n_1 + n_2)$.
 - U_1, U_2 : Pairwise comparison counts of how often one group exceeds the other.
 - U : Final test statistic evaluated against critical values of the Mann-Whitney

distribution.

- **What I Understood (Core Concept):** When data contains heavy skewness or severe outliers (e.g., salary data, Likert scales), means become distorted. Mann-Whitney bypasses raw values entirely, pools all data, and ranks them in order. It checks if the sum of ranks is evenly distributed or if one group consistently claims the higher ranks.

6. Chi-Square Test of Independence (χ^2)

- **Plain-English Definition:** A non-parametric test used with contingency tables to determine whether two categorical variables are statistically independent or significantly associated.
- **Formulas:**

$$\chi^2 = \sum \sum [(O_{ij} - E_{ij})^2 / E_{ij}]$$

$$E_{ij} = (R_i * C_j) / N$$

$$\text{Degrees of Freedom: } df = (r - 1) * (c - 1)$$

- **What Each Term Represents:**
 - **O_{ij}:** Observed frequency count in row i, column j.
 - **E_{ij}:** Expected frequency count in row i, column j under the assumption of independence.
 - **R_i, C_j:** Marginal totals for row i and column j.
 - **N:** Grand total of all observations.
 - **r, c:** Number of rows and columns in the contingency table.
- **What I Understood (Core Concept):** It asks whether membership in one category influences membership in another. You calculate the distribution expected if no relationship existed, calculate the squared difference between observed and expected counts, and normalize by expected counts. A large χ^2 statistic yields a small p-value, indicating an association between the variables.

7. One-Way ANOVA (Analysis of Variance)

- **Plain-English Definition:** An omnibus parametric test comparing the means of three or more independent groups by partitioning overall variance into between-group and within-group components.
- **Formulas:**

$$F = MS_{\text{between}} / MS_{\text{within}} = [SS_{\text{between}} / (k - 1)] / [SS_{\text{within}} / (N - k)]$$

$$SS_{\text{between}} = \sum n_i * (\bar{Y}_i - \bar{Y})^2$$

$$SS_{\text{within}} = \sum \sum (Y_{ij} - \bar{Y}_i)^2$$

- **What Each Term Represents:**
 - **F:** Variance ratio test statistic.
 - **SS_between / SS_within:** Sum of squares between groups (explained) and within groups (error/unexplained).
 - **MS_between / MS_within:** Mean squares (sum of squares divided by their respective degrees of freedom).
 - $\bar{Y}_i$, $\bar{Y}$: Group mean and grand overall mean.
 - **k:** Number of groups; **N:** Total observations.
- **What I Understood (Core Concept):** Running multiple pairwise t-tests inflates the overall Type I error rate (false positive risk). ANOVA avoids this by executing a single test that contrasts variance between group averages against natural variance within groups. An F-ratio significantly greater than 1 indicates at least one group mean differs from the rest, prompting a post-hoc test (such as Tukey's HSD) to pinpoint differences.

8. Multiple Linear Regression Model

- **Model Equation:**

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon$$

- **What Each Term Represents:**
 - **Y:** Continuous response/dependent variable.
 - β_0 : Intercept; expected value of Y when all predictors $X_j = 0$.
 - β_j : Partial regression slope; expected change in Y per 1-unit increase in X_j , holding all other predictors constant.
 - X_j : Explanatory/independent predictor variable.
 - ϵ : Random error term, assumed $\epsilon \sim N(0, \sigma^2)$.
 - $\hat{Y}$: Fitted/predicted response: $\hat{Y} = \beta_0 + \sum \beta_j X_j$.
 - e_i : Residual error for observation i: $e_i = Y_i - \hat{Y}_i$.
- **What I Understood (Core Concept):** Multiple regression isolates the unique influence of each predictor while holding confounding factors constant (ceteris paribus). It allows you to examine individual variables' relationships with the outcome while mathematically controlling for other variables in the model.

9. Model Metrics & Diagnostic Calculations

- **Formulas:**

$$t_statistic\ for\ \beta_j: t = \beta_j / SE(\beta_j)$$

$$95\%\ Confidence\ Interval: CI_{95\%} = \beta_j \pm (t_{crit} * SE(\beta_j))$$

$$R^2 = 1 - [SS_{res} / SS_{tot}] = 1 - [\sum (Y_i - \hat{Y}_i)^2 / \sum (Y_i - \bar{Y})^2]$$

$$\text{Adjusted } R^2 = 1 - [(1 - R^2)(n - 1) / (n - k - 1)]$$

$$\text{Jarque-Bera Statistic: } JB = (n / 6) * [S^2 + ((K - 3)^2 / 4)]$$

$$\text{Variance Inflation Factor: } VIF_j = 1 / (1 - R_j^2)$$

- **What Each Term Represents:**

- **β_j , $SE(\beta_j)$:** Estimated slope coefficient and its standard error.
- **t_{crit} :** Critical value cutting off 2.5% in each tail for $(n - k - 1)$ degrees of freedom.
- **SS_{res} / SS_{tot} :** Residual sum of squares (unexplained) and total sum of squares (total variance in Y).
- **n , k :** Sample size and number of predictors.
- **S , K :** Sample skewness and sample kurtosis of model residuals.
- **R_j^2 :** R^2 from regressing predictor X_j on all other $(k - 1)$ predictors.

- **What I Understood (Core Concept):**

- **Coefficients & p-values:** If a coefficient's 95% confidence interval contains zero, we fail to reject $\beta_j = 0$, meaning the variable does not have a statistically significant linear effect.
- **R^2 vs. Adjusted R^2 :** Standard R^2 increases whenever any predictor is added, regardless of usefulness. Adjusted R^2 incorporates a penalty based on degrees of freedom; it increases only if a new variable improves the model beyond what chance would predict.
- **Linearity & Homoscedasticity (Residuals vs. Fitted Plot):** Points should form a random horizontal band centered around zero. Systematic curves indicate non-linearity, while a funnel or fan shape indicates heteroscedasticity (non-constant variance), invalidating standard errors.
- **Normality of Residuals (Q-Q Plot / Jarque-Bera):** Residuals must be normally distributed for valid inference. On a Q-Q plot, points should follow the 45-degree diagonal. In the Jarque-Bera test, a p-value > 0.05 confirms residuals do not deviate significantly from normality.
- **Multicollinearity (VIF):** Predictors should explain the response, not duplicate each other. When $VIF > 5$ or 10 , strong collinearity inflates standard errors, making coefficient estimates unstable and p-values unreliable.

Summary Reference Table

Concept / Test	Primary Purpose	Key Assumption / Formula	Decision Criterion
Shapiro-Wilk	Normality check	$W = [\sum a_i x_{(i)}]^2 / \sum (x_i - \bar{x})^2$	$p > 0.05$ indicates normal distribution

Concept / Test	Primary Purpose	Key Assumption / Formula	Decision Criterion
Levene's Test	Equal variances check	ANOVA on absolute deviations $ Y_{ij} - \bar{Y}_i $	$p > 0.05$ indicates equal variances
Two-Sample t-Test	Compare 2 independent means	$t = (\bar{X}_1 - \bar{X}_2) / SE_{diff}$	$ t \geq \text{critical value}$ or $p \leq \alpha$
Mann-Whitney U	Non-parametric 2-group comparison	Rank-sum based pairwise comparison	$U \leq \text{critical value}$ or $p \leq \alpha$
Chi-Square (χ^2)	Independence of categorical variables	$\chi^2 = \sum [(O - E)^2 / E]$	$p \leq \alpha$ rejects independence
One-Way ANOVA	Compare 3+ group means	$F = MS_{between} / MS_{within}$	$F \geq \text{critical value}$ indicates ≥ 1 group differs
Multiple Regression	Model continuous outcome with multiple predictors	$Y = \beta_0 + \sum \beta_j X_j + \varepsilon$	Evaluate individual β_j p-values & overall fit
VIF	Check multicollinearity	$VIF = 1 / (1 - R_j^2)$	$VIF < 5$ is acceptable; $> 5-10$ indicates severe collinearity